INTEGRATION

BLAKE FARMAN

Lafayette College

Name: Solutions

1. Use right endpoints to evaluate the integral

$$\int_0^3 x^3 \, \mathrm{d}x.$$

Do **not** simply use the Fundamental Theorem of Calculus.

$f(x) = x^3$, $\Delta x = \frac{3-0}{n} = 3/n$, $x_i = 0 + i\Delta x = i\Delta x = \frac{3i}{n}$

$$\int_0^3 x^3 dx = \lim_{n\to\infty} R_n = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{3i}{n}\right)^3 \left(\frac{3}{n}\right) = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{81 i^3}{n^4}$$

$$= \lim_{n\to\infty} \frac{81}{n^4} \sum_{i=1}^{n} i^3 = \lim_{n\to\infty} \frac{81}{n^4} \left(\frac{n(n+1)}{2}\right)^2 = \frac{81}{4} \lim_{n\to\infty} \frac{(n^2+n)^2}{n^4}$$

$$= \frac{81}{4} \lim_{n\to\infty} \frac{(n^2)^2 + 2(n^2)(n) + n^2}{n^4} = \frac{81}{4} \lim_{n\to\infty} \frac{n^4 + 2n^3 + n^2}{n^4}$$

$$= \frac{81}{4} \lim_{n\to\infty} \frac{n^4}{n^4}\left(1 + 2/n + 1/n^2\right) = \frac{81}{4}(1 + 0 + 0) = \boxed{\frac{81}{4}}$$

Check: FTC $\Rightarrow$ $\int_0^3 x^3 dx = \frac{1}{4}x^4 \Big|_0^3 = \frac{1}{4}(3^4 - 0^4) = \frac{1}{4}(81 - 0) = \frac{81}{4}$ ✓

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Use the Fundamental Theorem of Calculus to evaluate the following definite integrals.

2. $\int_1^3 \left(x^2 + 2x - 4\right) \mathrm{d}x$

$$\int_1^3 (x^2+2x-4)\,dx = \int_1^3 x^2\,dx + 2\int_1^3 x\,dx - 4\int_1^3 dx$$
$$= \frac{1}{3}x^3\Big|_1^3 + x^2\Big|_1^3 - 4x\Big|_1^3$$
$$= \frac{1}{3}(3^3-1) + (3^2-1) - 4(3-1)$$
$$= \frac{26}{3} + 8 - 8$$
$$= \boxed{\frac{26}{3}}$$

3. $\int_0^1 \left(1 - 8v^3 + 16v^7\right) \mathrm{d}\cancel{v}\ dv$

$$\int_0^1 (1-8v^3+16v^7)\,dx = \int_0^1 dv - 8\int_0^1 v^3\,dv + 16\int_0^1 v^7\,dv$$
$$= v\Big|_0^1 - 8\left(\tfrac{1}{4}\right)v^4\Big|_0^1 + 16\left(\tfrac{1}{8}\right)v^8\Big|_0^1$$
$$= (1-0) - 2(1-0) + 2(1-0)$$
$$= 1 - 2 + 2 = \boxed{1}$$

4. $\int_1^8 x^{-2/3}\,\mathrm{d}x$

$$\int_1^8 x^{-2/3}\,dx = 3x^{1/3}\Big|_1^8$$
$$= 3\left(\sqrt[3]{8} - \sqrt[3]{1}\right)$$
$$= 3(2-1)$$
$$= \boxed{3}$$

5. $\int_{\pi/6}^{\pi/2} \csc(t)\cot(t)\,\mathrm{d}t$

$$\int_{\pi/6}^{\pi/2} \csc(t)\cot(t) = -\csc(t)\Big|_{\pi/6}^{\pi/2}$$
$$= -\left(\csc(\pi/2) - \csc(\pi/6)\right)$$
$$= \csc(\pi/6) - \csc(\pi/2)$$
$$= \frac{1}{\sin(\pi/6)} - \frac{1}{\sin(\pi/2)}$$
$$= \frac{1}{1/2} - 1 = 2 - 1 = \boxed{1}$$

6. $\int_{\pi/4}^{\pi/3} \csc^2(\theta)\,\mathrm{d}\theta$

$$\int_{\pi/4}^{\pi/3} \csc^2(\theta)\,d\theta = -\cot(\theta)\Big|_{\pi/4}^{\pi/3}$$
$$= \cot(\pi/4) - \cot(\pi/3)$$
$$= 1 - \frac{\cos(\pi/3)}{\sin(\pi/3)}$$
$$= 1 - \frac{1/2}{\sqrt{3}/2}$$
$$= 1 - \frac{1}{2}\left(\frac{2}{\sqrt{3}}\right) = \boxed{1 - \frac{\sqrt{3}}{3}}$$

7. $\int_0^{\pi/4} \sec(\theta)\tan(\theta)\,\mathrm{d}\theta$ $= \sec(\theta)$

$$\int_0^{\pi/4} \sec(\theta)\tan(\theta)\,d\theta = \sec(\theta)\Big|_0^{\pi/4}$$
$$= \sec(\pi/4) - \sec(0)$$
$$= \frac{1}{\cos(\pi/4)} - \frac{1}{\cos(0)}$$
$$= \frac{1}{\frac{\sqrt{2}}{2}} - \frac{1}{1}$$
$$= \frac{2}{\sqrt{2}} - 1$$
$$= \frac{2}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}} - 1$$
$$= \frac{2\sqrt{2}}{2} - 1 = \boxed{\sqrt{2} - 1}$$

8. $\int_0^1 \cos(\pi t/2)\,\mathrm{d}t$

$u = \frac{\pi t}{2}$, $du = \frac{\pi}{2}dt \Rightarrow \frac{2du}{\pi} = dt$

$$\int_0^1 \cos\left(\frac{\pi t}{2}\right)dt = \int_{u(0)}^{u(1)} \cos(u)\left(\frac{2}{\pi}\right)du$$

$$= \frac{2}{\pi}\int_0^{\pi/2} \cos(u)du$$

$$= \frac{2}{\pi}\left[\sin(\pi/2) - \sin(0)\right]$$

$$= \frac{2}{\pi}(1-0) = \boxed{\frac{2}{\pi}}$$

9. $\int_0^1 (2t-1)^{50}\,\mathrm{d}t$

$u = 2t-1$

$du = 2dt \Rightarrow \frac{1}{2}du = dt$

$$\int_0^1 (2t-1)^{50}dt = \int_{u(0)}^{u(1)} u^{50}\frac{du}{2} = \frac{1}{2}\int_{-1}^{1} u^{50}du$$

$$= \frac{1}{2}\left(\frac{1}{51}\right)\left(1^{51} - (-1)^{51}\right)$$

$$= \frac{1}{2}\left(\frac{1}{51}\right)(1-(-1)) = \frac{1}{2}\left(\frac{1}{51}\right)(2)$$

$$= \boxed{1/51}$$

10. $\int_0^{\pi/6} \frac{\sin(t)}{\cos^2(t)}\,\mathrm{d}t$

$u = \cos(t)$, $du = -\sin(t)dt \Rightarrow -du = \sin(t)dt$

$$\int_0^{\pi/6} \frac{\sin(t)}{\cos^2(t)}dt = \int_{u(0)}^{u(\pi/6)} \frac{-du}{u^2} = -\int_1^{\frac{\sqrt{3}}{2}} u^{-2}du = \frac{-1}{-1}u^{-1}\Big|_1^{\sqrt{3}/2}$$

$$= \left[\left(\frac{\sqrt{3}}{2}\right)^{-1} - 1\right] = \frac{2}{\sqrt{3}} - 1$$

$$= \boxed{\frac{2\sqrt{3}}{3} - 1}$$

11. $\int_0^3 x\sqrt{9-x^2}\,\mathrm{d}x$

$u = 9-x^2$, $du = -2xdx \Rightarrow \frac{-1}{2}du = xdx$

$$\int_0^3 \sqrt{9-x^2}\,xdx = \int_{u(0)}^{u(3)} \sqrt{u}\left(\frac{-1}{2}du\right)$$

$$= \frac{-1}{2}\int_9^0 u^{1/2}du = -\frac{1}{2}\left(\frac{2}{3}\right)u^{3/2}\Big|^0$$

$$= \frac{-1}{3}\left(0 - 9^{3/2}\right) = \frac{3^3}{3} = \boxed{9}$$

12. $\int_0^{\pi/2} \cos(x)\sin(\sin(x))\,\mathrm{d}x$

$u = \sin(x)$, $du = \cos(x)dx$

$$\int_0^{\pi/2} \cos(x)\sin(\sin(x))dx = \int_{u(0)}^{u(\pi/2)} \sin(u)du$$

$$= \int_0^1 \sin(u)du$$

$$= -\cos(u)\Big|_0^1$$

$$= -(\cos(1) - \cos(0))$$

$$= \boxed{1 - \cos(1)}$$

13. $\int_{-\pi/3}^{\pi/3} x^4\sin(x)\,\mathrm{d}x$

$f(x) = x^4\sin(x)$, $f(-x) = (-x)^4\sin(-x)$

$= x^4(-\sin(x))$

$= -x^4\sin(x) = -f(x)$ is odd

Let $u = -x$, so $-du = dx$

$$\Rightarrow \int_{-\pi/3}^{0} f(x)dx = \int_{u(-\pi/3)}^{u(0)} f(-u)(-du)$$

$$= \int_{\pi/3}^{0} f(u)du$$

$$= -\int_0^{\pi/3} f(u)du = -\int_0^{\pi/3} f(x)dx$$

$$\Rightarrow \int_{-\pi/3}^{\pi/3} f(x)dx = \int_{-\pi/3}^{0} f(x)dx + \int_{\pi/3}^{0} f(x)dx = \boxed{0}$$